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Programme	B. Sc. Computer Science				
Course Title	Introduction to Artificial Intelligence & Machine Learning				
Type of Course	Major				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamental Mathematics Concepts: Sets 2. Fundamentals of Python Programming				
Course Summary	This course provides an introduction to the ideas, techniques, and applications of artificial intelligence (AI) is given in this course. The fundamentals of knowledge representation, machine learning, and problem solving will be taught to the students.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Differentiate various knowledge representation methods, AI operations, Machine learning approaches and real-world applications.	U	C	Instructor-created exams / Quiz
CO2	Master Problem-Solving Techniques (search algorithms, heuristic approaches, and informed search strategies). Analyse and evaluate its efficiency.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Investigate the properties and applications of various machine learning techniques	Ap	C	Seminar Presentation /
				Group Tutorial Work/ Viva Voce

CO4	Evaluate Artificial Intelligence Search algorithms and Machine learning approaches' efficiency.	U	C	Instructor-created exams / Home Assignments
CO5	Implement and analyse Machine learning algorithms to solve practical problems.	Ap	P	Writing assignments/ Exams
CO6	Apply Concepts in Real-World Projects	Ap	P	Case Study/ mini Project
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks (70)
I	Introduction to Artificial Intelligence & Problem Solving and Searching		15	20
	1	Introduction to AI – AI problems, AI Techniques	2	
	2	Various AI Domains (Introduction only)	1	
	3	Problem Solving Techniques - Search Algorithms, Knowledge representation and reasoning, constraint satisfaction problems, Game playing, Machine learning, Simulated Annealing (Concepts only)	3	
	4	Uninformed search algorithms (breadth-first, depth-first)	3	
	5	Informed search algorithms (A*, heuristic search- Generate and Test, Hill Climbing, Best First Search)	6	
II	Knowledge Representation & Reasoning		10	15
	6	Knowledge representation using Propositional & Predicate Logic	3	
	7	Semantic Networks & Frames	3	
	8	Rule based system & Introduction to Expert System (Concepts only)	2	
	9	Reasoning- Forward Vs Backward reasoning & logics for non-monotonic Reasoning	2	
III	Introduction to Neural Networks		8	15
	10	Introduction to Artificial Neural Network	1	
	11	Understanding Brain & Perceptron Model	1	
	12	Single Layer Perceptron Model & Learning in Single layer Perceptron Model	2	
	13	Multi-Layer Perceptron Model & Learning in Multi-layer Perceptron Model	2	

	14	Introduction to python packages- keras & sklearn	2	
IV		Machine Learning Fundamentals	12	20
	15	Introduction to Machine learning- Applications of Machine Learning	1	
	16	Supervised Machine learning- Classification & regression algorithms (Introduction: Linear Regression, Decision tree)	2	
	17	Unsupervised Machine Learning-Clustering & Dimensionality Reduction (Introduction: K means Clustering, PCA)	2	
	18	Reinforcement Learning: Elements of Reinforcement Learning	2	
	19	Feature Engineering & Feature Selection	2	
	20	Building a classification model by training with data	1	
	21	Classification model evaluation- Introduction to confusion matrix	1	
	22	Practical implementation to set up a machine learning model	1	
V		Hands-on Artificial Intelligence & Machine Learning using Python: Practical Applications, Case Study and Course Project	30	30
	1	Implement the following: 1. Search algorithms BFS DFS 2. Neural Network Building a single layer perceptron using Keras 3. Multi-layer Neural Network Setting up a multi-layer perceptron model 4. Supervised machine learning Linear regression Decision tree 5. Unsupervised machine learning K means clustering PCA 6. Feature Engineering Feature selection from a dataset	20	
	2	Case study – AI tools / Use of AI in any movie	3	
	3	Implementation of Comparison of any two machine learning algorithms on a dataset	7	

References

- Elaine Rich, Kevin Knight, Shivsankar B Nair, “Artificial Intelligence”, Third Edition, Tata McGraw Hill Publisher
- Tom M. Mitchell, Machine Learning, McGraw-Hill, 1st Ed.
- Ethem Alpaydin, Introduction to Machine Learning- 3rd Edition, PHI.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	1	1	1	2	1						

CO 2	2	1	2	3	2	2						
CO 3	2	1	2	3	2	3						
CO 4	3	-	1	2	-	-						
CO 5	1	-	2	3	3	3						
CO 6	2	-	3	3	3	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓	✓	✓
CO 6	✓	✓	✓	